



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO TVR FUELS STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

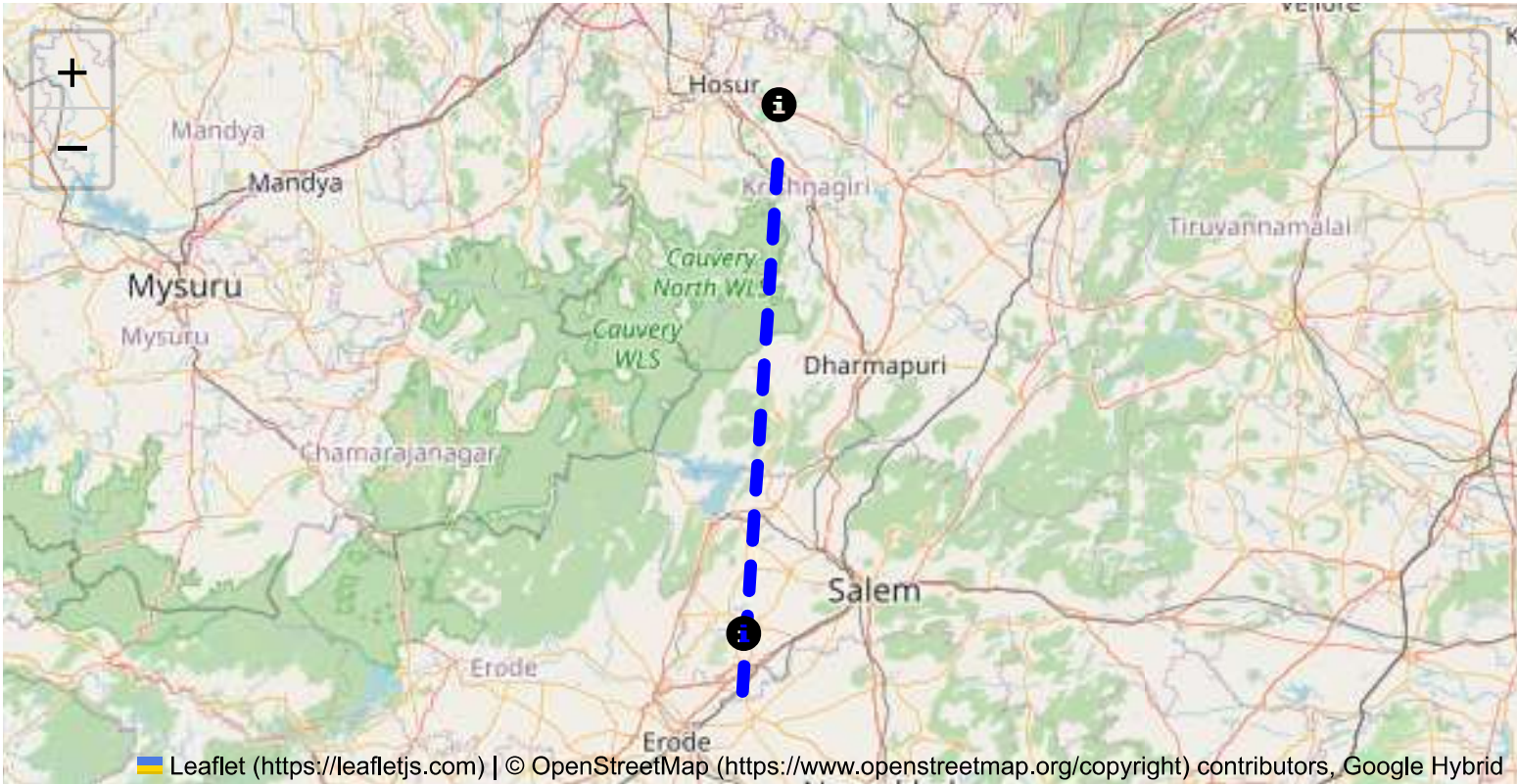
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 161.41 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.8 hours
Start: (11.4381, 77.8734)
End: (12.565749, 77.949409)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Ayaranapalli spans approximately 161.41 kilometers and follows a combination of national highways, state highways, and rural roads. Key areas along the route include the transition from National Highway 544 to State Highway 85, passing through towns like Tiruchengode and Krishnagiri, known for varying traffic and road conditions.

2. Typical Weather Conditions and Hazards

The region experiences a tropical climate with hot summers (March to June), monsoon rains (July to September), and mild winters (November to February). Weather-related hazards include:

- **Monsoon Season:** Heavy rains can lead to water-logged roads, reduced visibility, and potential landslides, especially in hilly areas near Krishnagiri.
- **Heat Waves:** During peak summer, extreme temperatures can affect vehicle performance, particularly increasing the risk of tire blowouts.

3. Analysis of Traffic Patterns

Traffic conditions vary along the route:

- **Peak Hours:** Morning (7-9 AM) and evening (5-8 PM) around major towns like Erode and Salem can experience heavy traffic congestion.
- **Congestion-Prone Areas:** Urban areas and junctions near Salem and Krishnagiri are common bottlenecks due to local traffic combined with long-haul vehicles.

4. Assessment of Road Quality and Infrastructure

- **Highways:** Generally well-maintained with good surface conditions. However, occasional construction work could cause temporary rough patches.
- **Rural Roads:** May have narrow lanes and less maintenance, resulting in potholes and uneven surfaces, particularly near small villages.

5. Alternative Routes for Emergencies

- **National Highway 44:** Can be used as an alternative for detours, connecting via Salem for a larger arc if SH 85 is inaccessible.
- **State Highway Detours:** Smaller state highways around Krishnagiri can serve as short emergency bypasses if the main route is blocked.

6. Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is subject to strict regulations including:

- **Permits and Documentation:** Required for transporting hazardous materials including route permits and emergency procedure documentation.
- **Time Restrictions:** Prohibited passage through certain areas during peak school and office traffic times for safety.

7. Overview of Historical Incidents

Historically, the route’s proximity to industrial areas sees occasional incidents involving chemical leaks or heavy vehicle accidents, often due to driver fatigue or mechanical failure.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** Pass through or near ecologically sensitive zones, particularly around forested regions near Krishnagiri.
- **Waste Management:** Must ensure no spillage or disposal of hazardous materials in water bodies along the route.

9. Communication Coverage

- **Urban Areas:** Good mobile network coverage due to proximity to towns and cities.
- **Rural Areas:** Potential dead zones particularly in forested or hilly areas; use GPS devices and radios if available.

10. Estimated Emergency Response Times

- **Urban Areas (Salem and Krishnagiri):** Approximately 30-45 minutes due to proximity to hospitals and emergency services.
- **Rural Areas:** Response times can exceed 60 minutes; prepare for delayed assistance.

11. Overall Summary of Risk Assessment

The route presents moderate risks, primarily from weather impacts during monsoons, potential traffic congestion around urban areas, and narrow roads in rural regions. While infrastructure generally supports heavy vehicles, adherence to regulations and vigilance around environmental protections are crucial. Communication infrastructure is variable, challenging coordination during emergencies in rural stretches. Preparing for varying conditions and ensuring thorough pre-trip checks and compliance with safety norms are recommended to minimize risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	High	11.47403, 77.86841	15 KM/Hr
6	Turn	High	11.49149, 77.89601	15 KM/Hr
7	Turn	High	11.49224, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.53501, 77.95707	30 KM/Hr
9	Turn	Medium	11.53601, 77.95856	30 KM/Hr
10	Turn	Medium	11.62403, 78.10893	30 KM/Hr
11	Turn	High	11.64180, 78.11808	15 KM/Hr
12	Turn	High	11.64325, 78.11993	15 KM/Hr
13	Turn	Medium	11.67865, 78.12803	30 KM/Hr
14	Turn	Medium	11.68201, 78.12639	30 KM/Hr
15	Turn	Medium	11.69519, 78.10775	30 KM/Hr
16	Turn	Medium	11.73958, 78.05105	30 KM/Hr
17	Turn	Medium	11.75835, 78.05016	30 KM/Hr
18	Turn	Medium	11.86847, 78.08367	30 KM/Hr
19	Turn	Medium	11.87942, 78.08197	30 KM/Hr
20	Turn	Medium	11.92585, 78.06507	30 KM/Hr
21	Turn	Medium	11.93704, 78.05402	30 KM/Hr
22	Turn	High	11.95818, 78.04978	15 KM/Hr
23	Turn	Medium	11.95988, 78.04948	30 KM/Hr
24	Turn	Medium	11.96304, 78.05136	30 KM/Hr
25	Turn	High	11.96399, 78.05135	15 KM/Hr
26	Turn	High	11.96580, 78.04923	15 KM/Hr
27	Turn	Medium	11.96675, 78.04939	30 KM/Hr
28	Turn	Medium	11.96791, 78.05070	30 KM/Hr
29	Turn	Medium	12.11983, 78.13276	30 KM/Hr
30	Turn	Medium	12.50505, 78.03133	30 KM/Hr
31	Blind Spot	Blind Spot	12.51384, 78.03205	10 KM/Hr
32	Turn	Medium	12.51395, 78.02896	30 KM/Hr
33	Turn	Medium	12.51637, 78.02568	30 KM/Hr
34	Turn	Medium	12.52171, 78.02417	30 KM/Hr
35	Turn	Medium	12.52712, 78.00951	30 KM/Hr
36	Turn	Medium	12.52900, 78.00856	30 KM/Hr
37	Turn	Medium	12.54948, 77.97277	30 KM/Hr
38	Turn	Medium	12.55078, 77.97257	30 KM/Hr
39	Turn	Medium	12.55126, 77.97216	30 KM/Hr
40	Turn	Medium	12.55317, 77.96530	30 KM/Hr
41	Turn	Medium	12.56517, 77.95315	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
14	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
20	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
21	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
25	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
26	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
27	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
28	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
31	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
32	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
33	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
34	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
35	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	college	Sona College	11.6768, 78.1257052	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
30	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
36	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
37	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64325, 78.11993



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67865, 78.12803



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68201, 78.12639



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69519, 78.10775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92585, 78.06507



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96399, 78.05135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11983, 78.13276



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50505, 78.03133



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.51384, 78.03205



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51395, 78.02896



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51637, 78.02568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52171, 78.02417



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52712, 78.00951



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52900, 78.00856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.54948, 77.97277



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55078, 77.97257



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55126, 77.97216



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55317, 77.96530



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56517, 77.95315

